# Social Influence Network Models for Opinion Formation and Decision-Making

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**Abstract**

This chapter provides an introduction to agent-based mathematical models of opinion formation and decision-making processes taking place on social influence networks. It is designed more as a tutorial to clarify key fundamental concepts, techniques, and models, rather than serve as a comprehensive account of the vast literature in this field. We begin by presenting the seminal DeGroot model, explaining its use in modelling opinion formation over networks, and state fundamental theoretical results. We then review several prominent models that extend the DeGroot model, showing how additional features can be introduced to capture additional social and psychological processes. Next, we provide examples of how these models can be employed, including problems dealing with the shaping of discussion outcomes, inferring the influence network structure, and experimental and observational studies utilising such models. A model of collective decision-making is presented, grounded in the theory of coordination games on networks, and we illustrate it's application to the study of innovation diffusion. Finally, we provide an extensive discussion on future research directions, including modelling, application, and deployment, as well as how such models could be applied to the study of greyzone environments and societal threats, in connection with the other chapters of the book.

# I. Introduction

Social network analysis (SNA) is an incredibly broad and rich field of research, and over the decades has seen significant contributions from a diverse range of research communities including sociology, psychology, economics, political science, physics, computer science and engineering (Freeman, 2004; Wasserman, Faust, et al., 1994). Problems in social network analysis may focus on study of the network structure itself to shed light on how communities and groups form and interact (Freeman, 2004; Wasserman, Faust, et al., 1994), or focus on modelling a *dynamical process* that occurs over a network structure to shed light on how local individual-level interactions can over time lead to emergent network-level phenomena (Barrat et al., 2008). This chapter deals with the latter, focusing specifically on how opinions form and evolve through social influence, which is a topic many now refer to as "opinion dynamics". Modelling how opinions within a social group form and evolve over time may provide an understanding of collective information processing and decision making in response to *societal threats* (see Chapter 1), and be used to examine the effects of a public information environment that is evolving with emerging digital technologies (see Chapter 13).

The field of opinion dynamics can be traced to John R. P. French Jr. and Bertram Raven's efforts to characterise *social power* in the 1950s. Indeed, French Jr (1956) proposed what is considered as the fundamental opinion dynamics model. The title of the paper and its narrative suggests that French was interested in identifying the outcome of a group discussion, including whether or not the group would reach a consensus of opinions, and which individual held the most social power, in the sense of having the most *influence* in determining the discussion outcome. Harary (1959) and DeGroot (1974) made seminal contributions to the analysis of the model after French published his work.

The French–Harary–DeGroot model considers a group of individuals discussing a particular topic or subject, with each individual holding an opinion[1]. Individuals share their opinions, and at discrete time instants, simultaneously update their opinions with an individual's new opinion being a weighted average of the others' current opinions. Weighted averaging is a mathematical implementation of *social influence*, which is a mechanism that causes individuals with differing opinions to try and resolve such differences by shifting their opinions to be closer together. Under certain conditions on the connectivity of the network structure describing the interactions of the group, a consensus of opinions is eventually reached, whereby all individuals hold the same opinion (DeGroot, 1974; French Jr, 1956; Harary, 1959).

[1] In the sequel, we provide both a mathematical representation of an opinion, and also discuss its general principle.

From these seminal works, the field of opinion dynamics emerged and grew, with thousands of papers now published on the subject[2]. A comprehensive taxonomy of the literature is a challenge beyond the scope of this chapter, but we attempt now to provide a simple characterisation to try and highlight the many different tools, methods and approaches employed. Papers may centre around the use of i) theoretical analysis tools, ii) computational methods and simulations, and/or iii) empirical approaches to study opinion dynamics. Theoretical tools can be used to identify conditions that yield convergence of the opinions to steady states (Altafini, 2013; M. Cao et al., 2008a; Parsegov et al., 2017), estimate the convergence rate (M. Cao et al., 2008b; Liu et al., 2017), characterise the steady state distribution of opinions (Parsegov et al., 2017; Ye, Liu, et al., 2020; Ye et al., 2019), and more. Computational and simulation tools allow for rapid study of more complex nonlinear models (Duggins, 2017; Mäs et al., 2014), or to explore specific case studies and scenarios that can involve thousands or hundreds of thousands of individuals (Nowak et al., 1990; Ye, 2019; Zino et al., 2020). Empirical methods may focus on validation of the model (De et al., 2016; N. E. Friedkin, 2006; N. E. Friedkin, Jia, & Bullo, 2016; Kozitsin, 2020), using it to explain particular social psychological phenomena (Becker et al., 2017), or making predictions (N. E. Friedkin & Bullo, 2017). Researchers may focusing on relaxing the assumptions imposed on the model while preserving the emergent phenomena, such as identifying a class of time-varying network structures that still guarantees a consensus of opinions is reached (M. Cao et al., 2008a). Perhaps most often, however, researchers explore the introduction of new individual-level mechanisms on top of the fundamental weighted averaging rule, and the new phenomena that consequently emerge at the network level. For instance, stubborn attachment to initial opinions can lead to a diverse disagreement (N. E. Friedkin & Johnsen, 1990), while antagonistic interactions may lead to polarisation (Altafini, 2013).

This chapter will focus on presenting the fundamentals of opinion formation and decision-making models, and important results. Throughout this chapter, we provide references for various results and problems, recognising that these are simply examples to serve as a starting point for readers, and that we may have excluded seminal and important contributions. We do this to keep our exposition clear and concise. For comprehensive accounts of the numerous contributions, readers may refer to the many excellent surveys, reviews and tutorial papers covering contributions across different scientific communities (Acemoglu & Ozdaglar, 2011; B. D. O. Anderson & Ye, 2019; Bernardo et al., 2024; Flache et al., 2017; Noorazar, 2020; Peyton Young, 2015; Proskurnikov & Tempo, 2017, 2018; Riehl et al., 2018). Finally, we conclude by noting that this chapter considers *agent-based models*, where actors in the population are represented by individual agents (sometimes referred to as microscopic models). In contrast, another broad area of the literature which are beyond the scope of this chapter, explores macroscopic models: the opinions of the population are described as a distribution, using methods from statistical and continuum mechanics (Düring & Wolfram, 2015; Mirtabatabaei & Bullo, 2012; Toscani, 2006; Toscani et al., 2018).

The rest of the chapter has the following structure. Section II presents the fundamental French–Harary–DeGroot model and key results, before covering important extensions and advances on the basic model. Section III provides a similar but shorter treatment of collective decision-making models, which are a natural companion of opinion formation models; social influence and information processing at first shapes an individual's opinion, and subsequently may result in particular actions or decisions being taken. Finally, Section IV provides a discussion of future research directions for this field, including relevance for addressing the societal issues explored in the rest of this book.

[2]As this chapter is being written, a quick search on Scopus reports 7,492 papers with the terms "opinion" and "dynamics".

## II. Models of Opinion Formation

In this section, we first present the French–Harary–DeGroot model and detail important conclusions that have been established in the literature regarding the model. For convenience, and as consistent with much of the literature, we henceforth call it the DeGroot model. For any theoretical results, no proofs are presented, and we point the interested reader to relevant texts. Then, we provide an overview of important contributions in the literature.

### A. The DeGroot Model

Consider a group of $n \geq 2$ individuals, who are indexed using the set $\mathcal{V} = \{1, \ldots, n\}$, interacting at discrete time instants $t = 0, 1, 2, \ldots$ to discuss opinions on a particular topic. For each individual $i \in \mathcal{V}$, we assume that they hold an opinion $x_i(t) \in \mathbb{R}$, which is a real-valued scalar number and $x_i(t)$ can change over time as individual $i$ interacts with other individuals and exchanges opinions. The DeGroot model posits that at each time instant $t$, the opinion of every individual $i \in \mathcal{V}$ evolves as

$$x_i(t+1) = \sum_{j=1}^{n} w_{ij} x_j(t), \tag{1}$$

with the nonnegative scalar $w_{ij} \geq 0$, $j \in \mathcal{V}$, being the *influence weight* that individual $i$ accords to individual $j$, and $\sum_{j=1}^{n} w_{ij} = 1$. How can we understand Eq. (1)?

First, let us clarify how $x_i$, which is a real-valued number, can represent an individual's "opinion". It is typical to assume that $x_i$ takes values from an interval between two real numbers, $a$ and $b$, $x_i \in [a, b]$, and then, we can think of $x_i$ as individual $i$'s cognitive orientation towards the particular object or topic under discussion with $x_i = a$ and $x_i = b$ representing extreme orientations. For instance

1) The group may be discussing COVID-19 vaccines, and $a = -1$ and $b = +1$. Then, $x_i = -1$ and $x_i = +1$ may indicate individual $i$ is fully against and fully supportive of COVID-19 vaccines, respectively. Meanwhile, $x_i = 0$ may indicate an individual is neutral. The magnitude of $x_i$ would indicate the intensity of the stance.
2) The group may be discussing the statement "The climate crisis is a human-made problem", and $a = 0$ and $b = 1$. Then $x_i = 0$ and $x_i = 1$ would represent an individual $i$ who is maximally certain that the statement is false and true, respectively, while values in between would indicate varying levels of certainty.
3) The group may be discussing the latest product from Apple, and with $a = -5$ and $b = +5$, an individual $i$ who extremely dislikes or extremely likes the product may have $x_i = -5$ and $x_i = +5$, respectively.

One can appreciate that the range of scenarios that the DeGroot model aims to capture is broad. The choice of $a$ and $b$ is not a critical issue, and the interval need not be symmetric about 0. What is important is that the interval $[a, b]$ is identified a priori and one must ensure that $x_i(t) \in [a, b]$ for all $t$ else the model is not well defined. As the following result states, this property indeed holds for the DeGroot model (and many other models, including those discussed later in this chapter).

**Proposition 1** (Ye (2019, Corollary 2.1))**.** *Consider a group of $n$ individuals, with opinions evolving according to Eq. (1). Then,* $\max_{j\in\mathcal{V}} x_j(0) \geq \max_{j\in\mathcal{V}} x_j(t)$ *and* $\min_{k\in\mathcal{V}} x_k(0) \leq \min_{k\in\mathcal{V}} x_j(t)$.

Thus, the opinions in the group never become more extreme than the most extreme initial opinions, and opinions are well defined for all time by setting $a \leq \min_{k\in\mathcal{V}} x_k(0)$ and $b \geq \max_{j\in\mathcal{V}} x_j(0)$.

Second, the requirement that $\sum_{j=1}^{n} w_{ij} = 1$ intimates that individual $i$'s opinion at the next time step, $x_i(t+1)$, is a convex combination or *weighted average*[3] of all the current opinions in the group, $x_j(t)$, with individual $j$'s opinion given the weighting $w_{ij} \geq 0$. This weighted averaging mechanism causes the $x_i$s to come closer together (or at least not become further apart), and it is a fundamental but abstract representation of the *social influence* that encourages two disagreeing individuals to resolve differences in opinions, or for an individual to become more similar to the rest of the group.

Last, we point out that $w_{ij} = 0$ for $j \neq i$ and $w_{ii} > 0$ are not only permitted within the DeGroot model but in fact often encountered. For the former, this implies that individual $j$ has no influence on individual $i$ during the group discussion. This might represent two individuals who are not able to communicate, e.g. a low-level employee never gets the chance to speak with the CEO of the company, or individual $i$ can receive individual $j$'s opinion but chooses to ignore it, e.g. the CEO chooses to ignore the employee's suggestion for a pay rise. This introduces the *network* aspect to opinion formation models, which we explore next. For the latter, $w_{ii}$ can be considered as individual $i$'s *self-confidence*, because it represents how much the individual considers others' opinions relative to their own. To see this, simply set $w_{ii} = 1$, which yields $w_{ij} = 0$ for all $j \neq i$ because we require $\sum_{j=1}^{n} w_{ij} = 1$.

### B. Fundamental Results of the DeGroot Model

We first introduce the notion of a *directed graph*, which is a tuple $\mathcal{G} = (\mathcal{V}, \mathcal{E}, W)$. The node set $\mathcal{V} = \{1, \ldots, n\}$ maps conveniently to the set of individuals in the group, so that individual $i$ corresponds to node $i$. The edges of the graph are captured by the set $\mathcal{E} \subset \mathcal{V} \times V$, and the edge $(j, i) \in \mathcal{E}$ represents an arc that starts at node $j$ and ends at node $i$. We say that $j$ is a neighbour of $i$. The adjacency matrix $W = (w_{ij})$ is a square matrix of dimension $n$ with nonnegative entries $w_{ij}$. More precisely, $w_{ij} > 0$ if and only if $(j, i) \in \mathcal{E}$. We say a graph is *strongly connected* if and only if one can start from any node $i$ and reach any node $j$ by following the edges while respecting the direction of each edge (Godsil & Royle, 2001).

Referring to Figure 1, the network view of the DeGroot model can therefore be considered by collecting the influence weights $w_{ij}$ in Eq. (1) to form the *influence* matrix $W$, and then associating with $W$ the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, W)$. When $w_{ij} > 0$, individual $j$ can *directly* influence individual $i$ during the opinion formation process. If one can reach individual $i$ from individual $j$ by following the edges, then individual $j$ *indirectly* influences individual $i$ through the others in the network. A strongly connected network thus is equivalent to every individual being able to (in)directly influence every other individual. If we collect the opinions of all individuals into a vector $x = [x_1, \ldots, x_n]^\top$, then the evolution of the opinions over time can be represented as the following linear time-invariant system

$$x(t+1) = Wx(t). \tag{2}$$

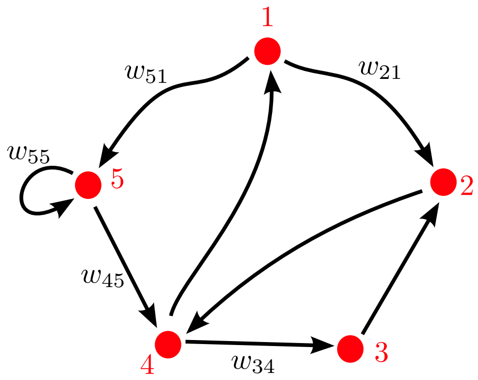
The following fundamental result establishes the outcome of the DeGroot opinion formation process on strongly connected networks.

**Theorem 1.** *Consider the system in Eq. (2) and suppose that $\mathcal{G}$ is strongly connected. Suppose further that there exists at least one individual $i \in \mathcal{V}$ with $w_{ii} > 0$. Then,*

$$\lim_{t \to \infty} x(t) = \alpha \mathbf{1}_n, \tag{3}$$

*at an exponentially fast rate, where $\alpha \in \mathbb{R}$ and $\mathbf{1}_n$ is the $n$-dimensional column vector of* 1*s.*

[3] To be precise, we are considering the weighted arithmetic mean. Other averages can also be considered; a weighted median mechanism will be introduced in Section III for decision-making.




$$W = \begin{bmatrix} 0 & 0 & 0 & w_{14} & 0 \\ w_{21} & 0 & w_{23} & 0 & 0 \\ 0 & 0 & 0 & w_{34} & 0 \\ 0 & w_{42} & 0 & 0 & w_{45} \\ w_{51} & 0 & 0 & 0 & w_{55} \end{bmatrix}$$

Fig. 1: An example of a strongly connected social network $\mathcal{G}$ and associated influence matrix $W$. Only some edge weights are labelled, and the $W$ matrix identifies all nonzero entries. Individual 1 directly influences individual 2, with weight $w_{21}$. Individual 5 has self-confidence $w_{55}$, and individual 1 can indirectly influence individual 3 via the path of edges $(1, 5), (5, 4), (4, 3)$.

This result can be recovered from Proskurnikov and Tempo (2017, Theorem 12), and states that for the group of individuals engaging in a discussion, their opinions will converge to a common value of $\alpha$ over time, and we know that $\alpha \in [\min_j x_j(0), \max_k x_k(0)]$ due to Proposition 1. That is, a *consensus* of opinions is reached, and such a consensus occurs irrespective of the initial opinions of any individual, the specific network structure (other than the topological conditions stated), or the values of the influence weights $w_{ij}$. The DeGroot model thus establishes that as a consequence of social influence (realised through the weighted averaging mechanism) and strong connectivity of the network (every individual can directly or indirectly influence every other individual), a consensus of opinions is always guaranteed. A numerical toy example of the DeGroot model is presented in Figures 2a and 3.

Besides being able to predict that a consensus of opinions is assured, it would also be of interest to predict the value of $\alpha$, viz. the final opinion value that everyone agrees on. The following result provides an affirmative answer, requiring a concept from linear algebra. The matrix $W$ is a *row-stochastic* matrix (each row sum is equal to 1) and has an eigenvalue at 1 which is unique when $\mathcal{G}$ is strongly connected (Horn & Johnson, 2012). Associated with this unity eigenvalue is the left Perron–Frobenius eigenvector $\gamma^\top = [\gamma_1, \ldots, \gamma_n]$, i.e., $\gamma^\top W = \gamma^\top$. This vector has all positive entries $\gamma_i > 0$ and the entries sum to 1, viz. $\sum_{i=1}^n \gamma_i = 1$.

**Corollary 1** (Proskurnikov and Tempo (2017, Section 3.5))**.** *In Theorem 1, the consensus opinion value is given by* $\alpha = \gamma^\top x(0)$.

Thus, the final consensus opinion $\alpha$ is a convex combination, or weighted average, of the initial opinions of the individuals in the group. Since $\sum_i \gamma_i = 1$, then the value $\gamma_i$ can be viewed as the contribution of individual $i$ in shaping the discussion outcome, relative to all other individuals. It is notable that *every* individual has a contribution, since $\gamma_i > 0$ for all $i$.

The theoretical analysis of the DeGroot model, represented in Eq. (2), is extremely mature (see Proskurnikov and Tempo (2017) for a comprehensive treatment with proofs). Here, we point out some important considerations. The first is that Theorem 1 is stated with a *topological* condition. That is, a certain connectivity property of the network $\mathcal{G}$ is required. Since the matrix $W$ is closely associated to $\mathcal{G}$, we remark that for every topological condition that guarantees a consensus of opinions is reached, there is a corresponding equivalent *algebraic* condition on $W$. The second is that strong connectivity of $\mathcal{G}$ and requiring at least one individual to have self-confidence is a *sufficient* condition for consensus to occur. There are more general sufficient conditions that drop the strong connectivity requirement, and there are yet more general conditions that are in fact *necessary and sufficient* for consensus. Once again, such conditions can be expressed in terms of the network topology $\mathcal{G}$ or the influence matrix $W$. In Theorem 1, an "exponentially fast" convergence rate is reported. This refers to the fact that there exist positive constants $k$ and $l$ such that

$$\|x(t) - \alpha\mathbf{1}_n\| \leq l\|x(0) - \alpha\mathbf{1}_n\|^{-kt} \tag{4}$$

for all $t \geq 0$, with $\|\cdot\|$ being a vector norm. The constant $k$ is of most interest, as it provides a lower-bound on the convergence rate, and it can be computed or estimated based on information on the $W$ matrix or the network topology structure, e.g. (Nedić & Liu, 2017; Olshevsky & Tsitsiklis, 2011).

Finally, we point out that Abelson (1964) proposed a *continuous-time* version of the DeGroot model as far back as 1964, with Eq. (1) being replaced by the ordinary differential equation

$$\dot{x}_i(t) = \sum_{j=1, j\neq i}^{n} a_{ij}\Big(x_j(t) - x_i(t)\Big), \tag{5}$$

where $a_{ij} \geq 0$ is the influence weight[4]. Importantly, there is no assumption on the value of the sum of $a_{ij}$, unlike the DeGroot model which states that $\sum_{j=1}^{n} w_{ij} = 1$. See Proskurnikov and Tempo (2017, Section 4.1) on how to relate the DeGroot and Abelson models. Just as in the discrete-time case, there are both algebraic and topological conditions for consensus of the Abelson model (Proskurnikov & Tempo, 2017; Ren & Beard, 2005). In the rest of this chapter, we focus on discrete-time models, because they are more widely adopted across different scientific communities working on opinion formation modelling (this may be due to their being easier to code and numerically simulate). Nonetheless, for most of the subsequent discussed literature, there are continuous-time counterpart results.

### C. *Important Extensions in Opinion Formation Modelling*

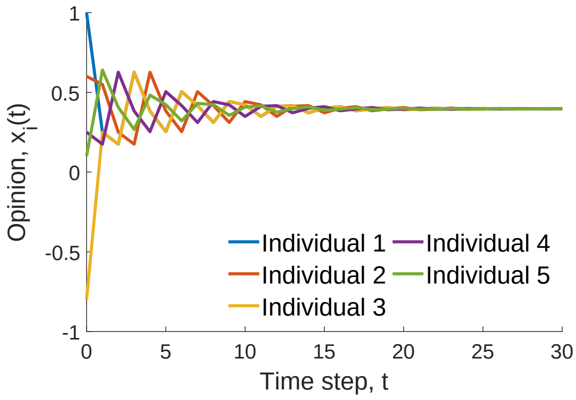
In the decades subsequent to the works of French, Harary, and DeGroot, there have been a number of important extensions to the fundamental weighted averaging DeGroot model. We explore a limited few, divided into themes covering a general mechanism or phenomenon being studied. For each of mechanism or phenomenon described below, there are works which consider a combination, amalgamation or variation; to ensure clarity, we focus on describing each separately.

*1) Interaction mechanisms:* The DeGroot model, captured by Eq. (2), assumes that every individual interacts and updates opinions *synchronously*, i.e. at the exact same time. Clearly this is an unrealistic assumption, because even in the same group, individuals may interact at different times and there is no guarantee that individuals will change opinions at the same rate. Synchronous updating also leads to counter-intuitive predictions in special cases. For example, consider two individuals with the following implementation of Eq. (2):

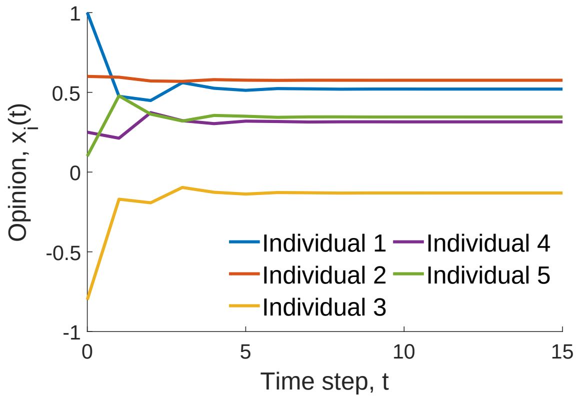
$$\begin{bmatrix} x_1(t+1) \\ x_2(t+1) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}. \tag{6}$$

Then, for any $x(0)$ such that $x_1(0) \neq x_2(0)$, a consensus can never be reached. In fact, a periodic trajectory emerges: $x(k) = [x_1(0), x_2(0)]^\top$ and $x(k+1) = [x_2(0), x_1(0)]^\top$ for $k = 0, 2, 4, 6 \ldots$. This periodic trajectory occurs even though (in fact precisely because) each individual places all of the influence weight on the other individual. Note that even though the graph is strongly connected, the $W$ in Eq. (6) does not satisfy Theorem 1 because $w_{ii} = 0$ for $i = 1, 2$. A number of different works aim to relax or

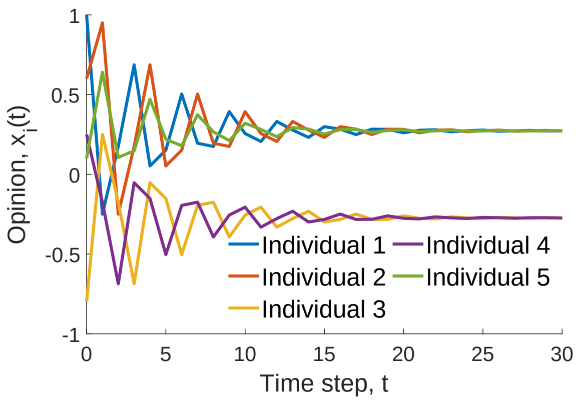
[4]Eq. (5) is also known as the linear single-integrator consensus algorithm. It has been the subject of extensive study in cooperative control and coordination of multi-agent and networked systems. Applications range from synchronisation of micro-grids, distributed optimisation, formation control of autonomous vehicles, and economic dispatch in power networks (Y. Cao et al., 2013; J. Qin et al., 2017; Ren & Cao, 2011).



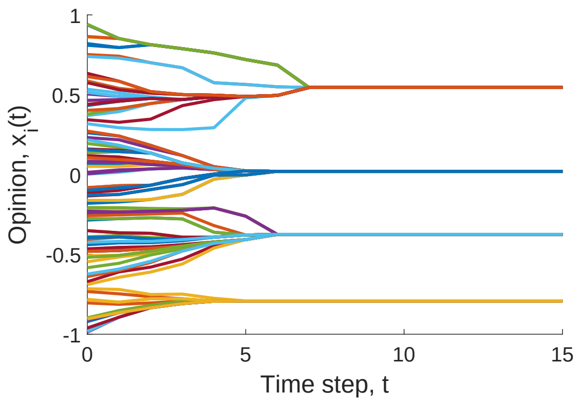

(a) DeGroot



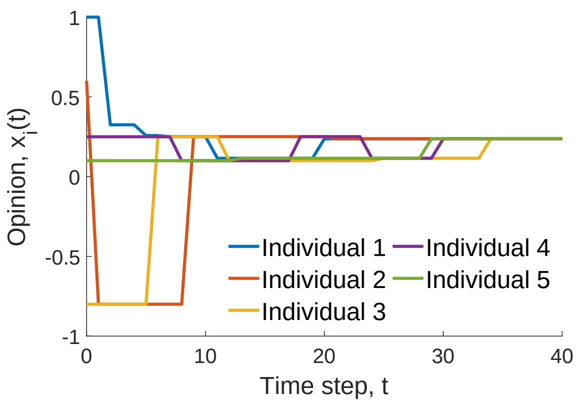

(b) Friedkin–Johnsen



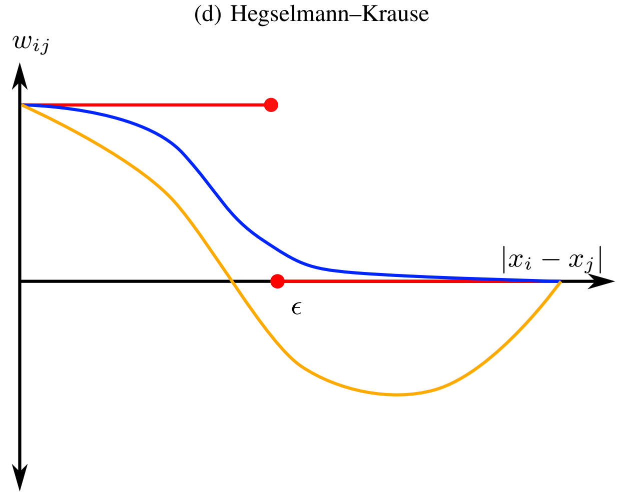

(c) Altafini




(d) Hegselmann–Krause

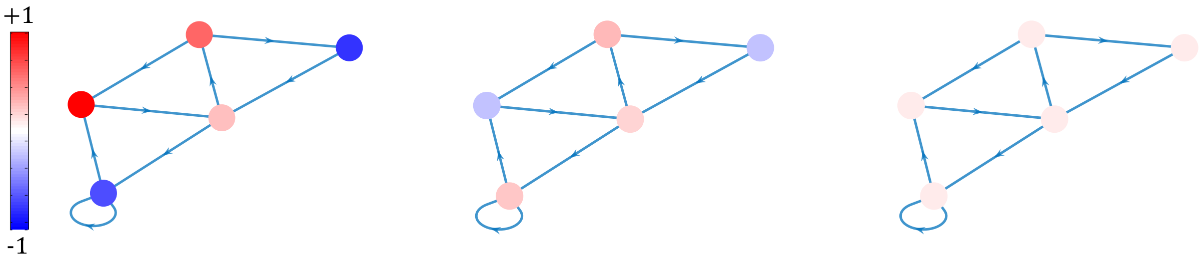



(e) DeGroot gossip



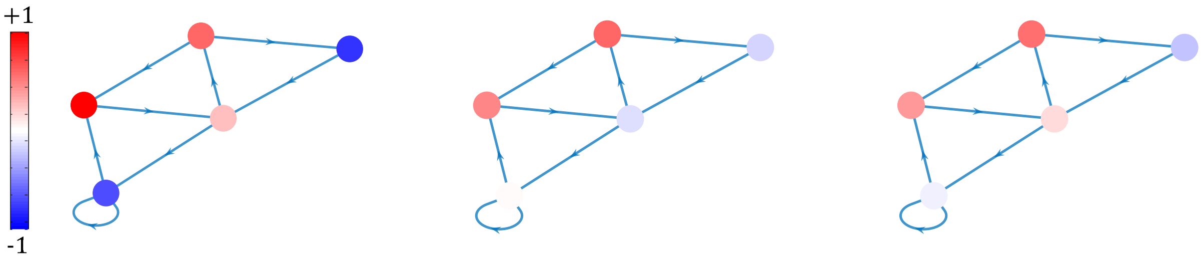

(f) Nonlinear influence weight functions

Fig. 2: Illustrative examples of different opinion formation models. (a) The DeGroot model leads to a consensus; (b) The Friedkin–Johnsen model predicts a persistent disagreement due to individual attachment to existing prejudice; (c) The Altafini model predicts modulus consensus due to antagonistic interactions in a structurally balanced network where opinion values have equal magnitude but can have opposite signs. (d) The Hegselmann–Krause model with bounded-confidence leads to cluster consensus, where each opinions within each cluster are the same but different between clusters. (e) Gossip-based DeGroot model still leads to a consensus but the consensus value differs from (a) and is dependent on the activation sequence of individuals. (f) Examples of nonlinear influence weight functions $w_{ij}(|x_i-x_j|)$: the standard bounded-confidence function with confidence parameter $\epsilon$ (red), positive influence that decreases with opinion distance (blue) and influence that is positive or negative depending on opinion distance (orange).



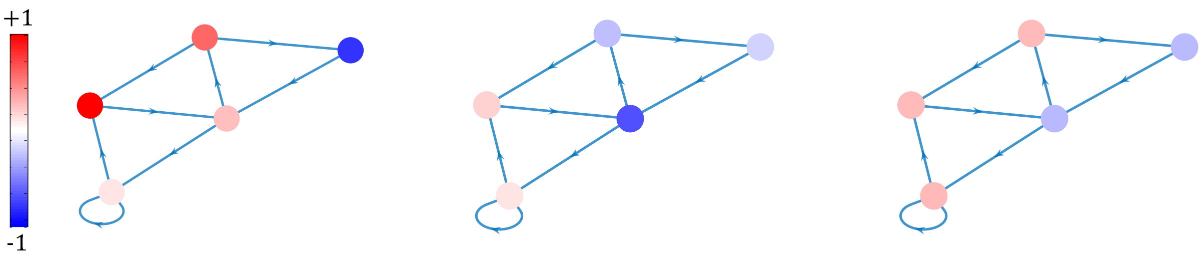

(a) $t = 0$ (b) $t = 5$ (c) $t = 30$

Fig. 3: A network perspective of the DeGroot model. The colour and intensity of each node reflects the opinion held by the individual at time $t$ for (a) $t = 0$, (b) $t = 5$, and (c) $t = 30$. Over time, a consensus (same colour and intensity) is reached, but in the transient, individuals opinions can evolve (including into the opposite colour).




(a) $t = 0$ (b) $t = 1$ (c) $t = 30$

Fig. 4: A network perspective of the Friedkin–Johnsen model. The colour and intensity of each node reflects the opinion held by the individual at time $t$ for (a) $t = 0$, (b) $t = 1$, and (c) $t = 30$. Due to stubborn attachment to existing prejudices, no consensus is reached, and persistent disagreement remains.




(a) $t = 0$ (b) $t = 4$ (c) $t = 30$

Fig. 5: A network perspective of the Altafini model. The colour and intensity of each node reflects the opinion held by the individual at time $t$ for (a) $t = 0$, (b) $t = 4$, and (c) $t = 30$. Due to the antagonistic interactions and a structurally balanced network, the opinions split into two clusters of individuals: opinions have equal magnitude but opposite sign between the clusters.

adjust the assumptions around the interaction mechanisms, by introducing *asynchronous* activation and updating.

One mechanism which has received significant attention is the *gossip* mechanism (Acemoglu & Ozdaglar, 2011; Boyd et al., 2006; Liu et al., 2011; Proskurnikov & Tempo, 2018; Ravazzi et al., 2015). The basic idea is that at each time $t$, two individuals exchange opinions, while other individuals remain inactive. Different assumptions can be considered in determining which two individuals activate at time $t$. One popular implementation begins by assuming there is an underlying *backbone* network $\mathcal{G}$ capturing all the possible interactions between individuals. At each time $t$, each individual has a positive probability of being activated, independent of past activation events. The active individual $i$ contacts one of their neighbours in the backbone network, with each neighbour having a positive probability of being selected. An illustrative example is presented in Figure 6. If the active individual $i$ contacts individual $j$, then

$$x_i(t+1) = \zeta_i x_i(t) + (1 - \zeta_i)x_j(t) \tag{7}$$

and $x_k(t+1) = x_k(t)$ for all $k \neq i$. The parameter $\zeta_i \in (0, 1)$ can be considered the self-confidence. Then, consensus of opinions is achieved with probability 1 if $\zeta_i \in (0, 1)$ for all $i \in \mathcal{V}$ (Fagnani & Zampieri, 2008; Proskurnikov & Tempo, 2018), with Figure 2e providing an illustrative example. On the other hand, if there are stubborn individuals, with $\zeta_i = 1$, then the opinions can fluctuate constantly over time, but the *expected values* of the opinions do converge (Acemoğlu et al., 2013; Parsegov et al., 2017; Ravazzi et al., 2015). In other implementations of the gossip mechanism, two individuals activate, interact, and then both update their opinions simultaneously (Boyd et al., 2006). Going further, one can also assume that the single active individual $i$ contacts all of their neighbours and enacts the regular DeGroot model in Eq. (1), with different assumptions on the how individuals are selected to become active (Y. Chen et al., 2019; Y. Qin et al., 2019; Xia et al., 2020; Zino et al., 2019). Asynchronous updating can resolve the oscillating behaviour detailed in Eq. (6), ensuring a consensus where synchronous updating would not (Y. Qin et al., 2019, Section 4).

Finally, we mention *time-varying* interactions, i.e., the network structure changes over time, as captured by the graph $\mathcal{G}(t) = (\mathcal{V}, \mathcal{E}(t), W(t))$. This may capture for instance friendships forming and ending over the course of the opinion formation process, or that at any one time, an individual may only discuss with a limited number of others but may change who they discuss with as time goes on. Some works derive sufficient conditions on the nature of the time-varying interactions that is possible while still preserving the outcome of consensus (Blondel et al., 2005; M. Cao et al., 2008a; Shi & Johansson, 2013; Xia et al., 2018, 2019). For example, the DeGroot model will still reach a consensus if the network is *jointly strongly connected*; roughly speaking, the union of edges over a sufficiently long interval must form a strongly connected network, and this must happen infinitely often.

*2) Susceptibility to social influence:* It was noticed early on following the developments of French that the weighted averaging mechanism, which aimed to capture social influence, would lead to a consensus for networks satisfying some mild connectivity assumptions. However, a consensus of opinions is often not achieved in the real world. In fact, one often sees a diverse range of opinions, including "community cleavage" in which a population is divided into two opposing partisan camps. Abelson's diversity puzzle summarises this neatly (Abelson, 1964, pg. 153) (paraphrased): if social influence models suggest that a consensus is the typical outcome of opinion formation, what other mechanisms must be at play to generate the diverse opinions observed in reality?

One intuitive mechanism is closure or reduced susceptibility to social influence, which might manifest in each individual as an attachment to existing prejudices and opinions. This was first proposed as a continuous-time model by Taylor (1968), but popularised by Noah E. Friedkin and Eugene Johnsen in

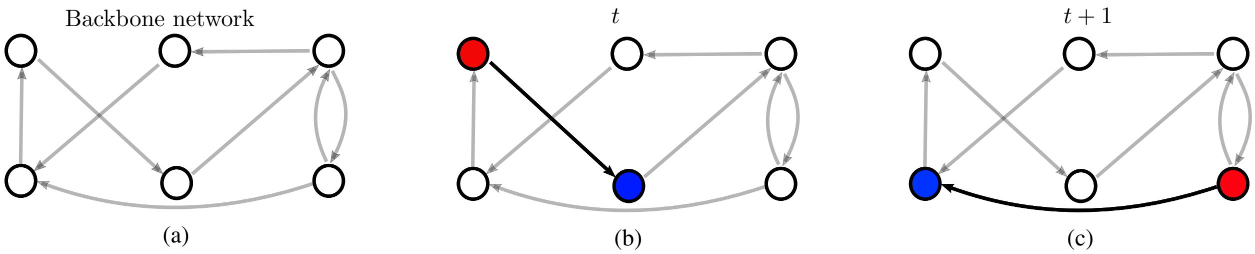



Fig. 6: Illustrative example of the gossip interaction mechanism. (a) The backbone network identifies all the possible interactions between individuals in the network. (b) At each time $t$, an individual $i$ is activated (red node) with uniform probability, contacts one of their neighbours $j$ (blue node) to learn of the neighbour's opinion, and then updates according to Eq. (7). (c) At the next time step, a different individual $k$ may activate (red node) and contact one of their neighbours (blue node).

the now seminal Friedkin–Johnsen (FJ) discrete-time model (N. E. Friedkin & Johnsen, 1990). In the FJ model, individual $i$'s opinion evolves as

$$x_i(t+1) = \lambda_i \sum_{j=1}^{n} w_{ij} x_j(t) + (1-\lambda_i) u_i, \tag{8}$$

where $\lambda_i \in [0,1]$ is individual $i$'s susceptibility to social influence, while $u_i \in \mathbb{R}$ is individual $i$'s existing constant prejudice. A common choice is $u_i = x_i(0)$, which means individual $i$ can become anchored to their initial opinion if $\lambda_i < 1$. Indeed, $x_i(t+1)$ becomes the convex combination of the current social influence and the existing prejudice, and $(1-\lambda_i)$ represents the individual's stubbornness to change. To see this, notice that $\lambda_i = 1$ and $\lambda_i = 0$ represents an individual who is maximally open and maximally closed to social influence, respectively, with the DeGroot model recovered when $\lambda_i = 1$.

Various different topological and algebraic conditions exist for the FJ model to converge (N. E. Friedkin & Johnsen, 1990; Ghaderi & Srikant, 2014; Parsegov et al., 2017; Proskurnikov & Tempo, 2017; Proskurnikov et al., 2017), and the typical outcome is that a diverse set of opinions is observed at steady state, and a consensus fails to be achieved. A toy numerical example of the FJ model demonstrating this diversity is given in Figures 2b and 4. The FJ model is notable because i) it is a simple and intuitive extension to the DeGroot model, ii) it can produce a broader range of opinion formation phenomena from the real world, and iii) it has had extensive empirical studies to explore its applicability (more on the last point in the sequel).

*3) Bounded confidence models:* "Birds of a feather flock together" is a popular proverb that encapsulates the sociological concept of "homophily" (McPherson et al., 2001) – the focus of many empirical studies in the SNA literature – which states that individuals tend to interact with, and have membership in the same groups as other like-minded individuals who share similar interests, opinions, experiences etc. In the context of this chapter, homophily can be studied using a broad class of models known as bounded confidence models (Bernardo et al., 2024; Lorenz, 2007).

The first such model was introduced in (Krause, 2000), became widely known through the seminal work of Hegselmann and Krause (2002), and is now known as the Hegselmann–Krause (HK) model. In the simplest implementation, the HK model extends the DeGroot model by positing that individual $i$'s opinion evolves as

$$x_i(t+1) = \frac{1}{\sum_{j=1}^{n} w_{ij}\big(x(t)\big)} \sum_{j=1}^{n} w_{ij}\big(x(t)\big) x_j(t), \qquad w_{ij}\big(x(t)\big) = \begin{cases} 1 & \text{if } |x_i(t) - x_j(t)| \leq \epsilon \\ 0 & \text{otherwise} \end{cases} \tag{9}$$

where $|\cdot|$ denotes the absolute value, and $\epsilon > 0$ is the confidence bound parameter. Individual $i$ will be influenced by individual $j$ if and only if the two individuals' opinions are similar enough, with similarity captured by the $\epsilon$ parameter. The first point to notice is that the HK model has *nonlinear* dynamics, in the sense that the update equation is nonlinear in the states $x_1(t), \ldots, x_n(t)$. The second remark is that under the HK dynamics in Eq. (9), the opinions will converge to a configuration known as *clustered consensus*; there may be one or more clusters of individuals where individuals within the same cluster have reached a consensus, but this consensus value is different between clusters. See Figure 2d for a numerical example.

Due to the strong nonlinearity, it is significantly more challenging to secure theoretical results, for bounded confidence models compared with the DeGroot or FJ models (Proskurnikov & Tempo, 2018). On the other hand, this nonlinearity allows the generation of more diverse and complex opinion formation phenomena, making it popular for studies driven by numerical simulations. Over the past two decades, the HK model has been extended in different directions (Bernardo et al., 2024; Lorenz, 2007), such as incorporating heterogeneous confidence bounds for different individuals, and asynchronous updating (the latter is in fact referred to as the Deffuant–Weibusch model (Deffuant et al., 2000)). Moreover, recent studies have considered a general function $w_{ij} = f_i(|x_i(t) - x_j(t)|)$, where $f_i$ is a nonnegative function that is monotonically decreasing in $|x_i(t) - x_j(t)|$ to capture the fact that the social influence individual $j$ exerts on individual $i$ may decrease as $j$'s opinion becomes less similar to individual $i$'s opinion (Duggins, 2017; Mäs et al., 2014). Figure 2f presents an example of the standard HK bounded-confidence (red) and a general function $f_i$ (blue).

*4) Antagonistic interactions:* The "backfire effect" refers to the possibility that, when an individual receives information from a distrusted or disliked source, the individual may reject the information and adopt an opposing position. This notion can be encapsulated in opinion formation models through antagonistic interactions, or negative influence (see Shi et al. (2019) for a recent review). The Altafini model (Altafini, 2013) extends the DeGroot model in Eq. (1) by allowing $w_{ij} < 0$ if individual $i$ views their relationship with individual $j$ to be antagonistic, competitive, or unfriendly. It is assumed that $w_{ii} \geq 0$, and $\sum_{j=1}^{n} |w_{ij}| = 1$ for all $i$.

In the context of the graph $\mathcal{G}$ representing the influence network structure, the introduction of negative $w_{ij}$ yields a *signed graph* with signed edges; the edge $(j, i) \in \mathcal{E}$ is said to be a negative or positive edge according as $w_{ij} < 0$ or $w_{ij} > 0$. Signed graphs have a multitude of applications in network science, and *structural balance* is a fundamental concept tracing back to the 1950s (Cartwright & Harary, 1956). Specifically, a signed graph is structurally balanced if and only if one can divide the set of nodes $\mathcal{V}$ into two disjoint subsets $\mathcal{V}_1$ and $\mathcal{V}_2$ with every edge $(i, j) \in \mathcal{E}$ having the following properties:

1) If $\{i, j\} \in \mathcal{V}_1$ or $\{i, j\} \in \mathcal{V}_2$, then $(i, j)$ is a positive edge;
2) If $i \in \mathcal{V}_k$ for $k \in \{1, 2\}$ and $j \notin \mathcal{V}_k$, then $(i, j)$ is a negative edge.

In other words, the network is split into two camps with all interactions between individuals in the same camp being positive/friendly, while interactions between individuals of different camps are negative/hostile[5]. A graph that is not structurally balanced is said to be *structurally unbalanced*. Figure 7 provides an illustrative example. An important result of the Altafini model ties structural balance to the emergence of polarised opinions as follows.

**Theorem 2.** *Consider the system in Eq. (2) and suppose that $\mathcal{G}$ is a strongly connected signed graph, with at least one individual $i \in \mathcal{V}$ with $w_{ii} > 0$. Then, the following hold:*

*1) If $\mathcal{G}$ is structurally balanced, $\lim_{t\to\infty} x(t) = \alpha \boldsymbol{p}$, where $\alpha \in \mathbb{R}$ and $\boldsymbol{p}$ is a vector with ith entry satisfying $p_i = +1$ if $i \in \mathcal{V}_1$ and $p_i = -1$ if $i \in \mathcal{V}_2$.*

[5]Indeed, structural balance generalises proverbs such as "an enemy of an enemy is my friend" and "a friend of my enemy is my enemy".

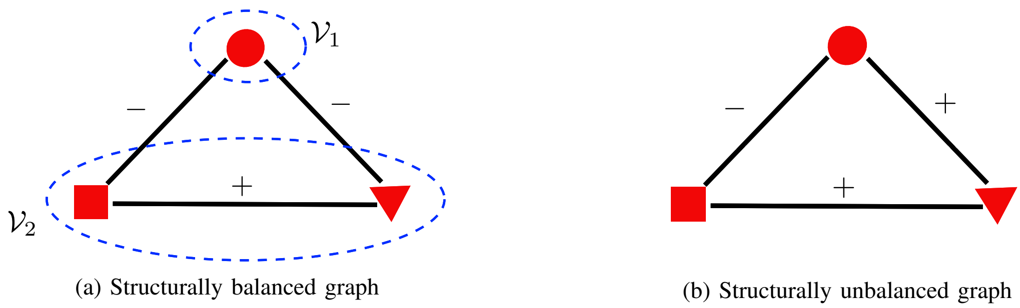



Fig. 7: Illustrative example of signed graphs, with positive and negative edges marked by $+$ and $-$, respectively. (a) This signed graph is structurally balanced, since we can divide the nodes into two camps of $\mathcal{V}_1 = \{\text{circle}\}$ and $\mathcal{V}_2 = \{\text{triangle, square}\}$. Edges have negative sign between camps, and positive sign within camp. From the square nodes perspective, this example reflects the proverb: "an enemy (triangle) of my enemy (circle) is my friend". (b) This signed graph is structurally unbalanced, which in social networks may result in tension and discomfort: square and circle dislike each other, but both are friends with triangle.

*2) If $\mathcal{G}$ is structurally unbalanced,* $\lim_{t\to\infty} x(t) = \mathbf{0}_n$, *where* $\mathbf{0}_n$ *is the $n$-dimensional vector of* $0$s.

If $\mathcal{G}$ is structurally balanced, then all opinions reach the same value in magnitude, $|x_i(\infty)| = |x_j(\infty)| = \alpha$ for all $i, j \in \mathcal{V}$ but the *sign* of the opinion depends on which of the two camps $\mathcal{V}_1$ and $\mathcal{V}_2$ the individual belongs to[6]. This is also referred to as modulus consensus or bipartite consensus, and an illustrative numerical example is presented in Figures 2c and 5. On the other hand, if $\mathcal{G}$ is not structurally balanced, then all opinions converge to $0$. From the perspective of the social network, and if for instance we assume opinions exist in the interval $[-1, +1]$, then this outcome can be contextualised as everyone reaching a consensus on the 'neutral' opinion – no biased subsets of individuals exist. We remark that Theorem 2 assumes the graph $\mathcal{G}$ is time-invariant, i.e., the edges and their signs do not change for all $t \geq 0$. Some results have established conditions for convergence under time-varying graphs (Liu et al., 2017; Shi et al., 2019), but these are topological conditions that do not posit a *mechanism* for how the edges and their signs change over time. The results discussed here need to be considered separate to (but perhaps in light of) a body of literature concerned with how signed graphs evolve. Such works typically begin by noting that unbalanced graphs are unstable[7], and tend towards a balanced state – various models have been proposed and studied to explain how unbalanced graphs eventually tend towards structural balance (De Pasquale & Valcher, 2022; Jia et al., 2016; Marvel et al., 2011; Mei, Chen, et al., 2022; Mei et al., 2019).

The notion of negative influence has been incorporated into bounded confidence models, whereby in Eq. (9), one has $w_{ij} = -1$ if $|x_i(t) - x_j(t)| > \hat{\epsilon}$ for some $\hat{\epsilon} > \epsilon > 0$, and we replace the denominator in Eq. (9) with $\sum |w_{ij}|$ (Aghbolagh et al., 2019, 2020). If a general function $f_i$ is considered as described in Section II-C3, then one can similarly extend to consider negative influence by allowing $f_i$ to be negative for sufficiently large values of $|x_i(t) - x_j(t)|$, as illustrated in Figure 2f (orange).

*5) Opinion formation over multiple topics:* So far, we have focused on social networks discussing a single topic and each individual holds one opinion value, $x_i(t)$. We conclude by introducing models for opinion formation over multiple topics, a problem which has recently received significant attention with with at least three different strands.

[6]An unsigned graph, e.g. a graph from the DeGroot model, is always structurally balanced since $\mathcal{V} = \mathcal{V}_1$ while $\mathcal{V}_2 = \emptyset$. Thus, the result in Theorem 2 subsumes the result in Theorem 1.

[7]Unstable in the sense there is discomfort and tension among the actors in the network (Cartwright & Harary, 1956), rather than unstable in the dynamical systems context (Sastry, 1999).

First, we can consider a scenario in which individuals are simultaneously discussing $m \geq 2$ topics. When the topics are independent from one another, the existing results from the DeGroot and FJ models extend immediately. On the other hand, topics may be interdependent: an individual's opinion on whether the 2003 US-led invasion of Iraq was a justified war likely depends on their opinion on whether Iraq had weapons of mass destruction in 2003 and intended to use them. A number of works have explored models of interdependent opinion formation, identifying that the interdependence structure itself plays an important role in shaping the discussion outcomes (Bizyaeva et al., 2022; Franci et al., 2023; N. E. Friedkin, Proskurnikov, et al., 2016; Kashima et al., 2021; Nedić et al., 2019; Parsegov et al., 2017; Ye, Liu, et al., 2020; Ye, Trinh, et al., 2020).

Second, one can consider individuals discussing a set of topics sequentially. For instance, a government cabinet may meet periodically to discuss economic, social security, defence and industry concerns, or a company board may discuss the same issue over several meetings to reach a resolution. Results have be established showing that, using the Friedkin–Johnsen model to capture the opinion dynamics, that discussion of the same topic repeatedly can produce consensus when a single discussion fails to do so (Bernardo et al., 2021; L. Wang et al., 2022). When discussing topics in a sequence, it is not just the opinion formation process for each topic that is of interest, but *appraisal dynamics* begin to play an important role. Roughly speaking, during group discussion, an individual's self-confidence $w_{ii}$ may increase or decrease as they observe and appraise of their success or failure in making an impact and persuading others. This leads to dynamic models describing how $w_{ii}$ changes over the sequence of topics[8], as opposed to how opinions $x_i$ change (B. D. O. Anderson & Ye, 2018; G. Chen et al., 2019; X. Chen et al., 2017; Jia et al., 2015, 2019; Tian et al., 2021; Ye & Anderson, 2019; Ye et al., 2018).

Finally, the notion that an individual does not necessarily share their true (private) opinion with others when engaging in group discussion or discourse, whether through a conscious choice or subconsciously, has a long history in the social psychology literature (Kuran, 1997; Noelle-Neumann, 1993; Prentice & Miller, 1993; Waters & Hans, 2009). Recent models have assumed that each individual has both a private opinion and a public (expressed) opinion, each evolving through coupled dynamics (Banisch & Olbrich, 2019; Duggins, 2017; Jadbabaie et al., 2022; Shang, 2019; Xia et al., 2020; Ye et al., 2019). These models describe individuals who may express an opinion that is different from their private opinion (Ye et al., 2019), which at a population level correspond to the emergence of interesting social psychological phenomena such as unpopular norms (Centola et al., 2005), pluralistic ignorance (Prentice & Miller, 1993), and the "spiral of silence" (Noelle-Neumann, 1993). For instance, pluralistic ignorance refers to a social group or community where the majority of its individuals privately hold a certain opinion, but incorrectly assume that the majority support a different opinion.

### *D. Employing Opinion Formation Models*

So far, we have detailed the original DeGroot model for opinion formation, and explored a number of important extensions in the modelling. Now, we identify several streams of literature that employ opinion formation models to study specific problems or scenarios. The number and nature of such problems are vast, and we only broadly cover three notable streams, viz. influence maximisation, zealots and stubborn extremists, and inference of network structure. We remark that such applications may be considered from the perspective of grey zone activities and information warfare, as highlighted in Chapters 1 and 14, whether to achieve a desired outcome through an information operation, or understand the impact of and then defend against malign foreign influence.

[8]Since $\sum w_{ij} = 1$, changes in $w_{ii}$ imply changes to $w_{ij}$ for $j \neq i$.

*1) Shaping the discussion outcomes:* A fundamental problem within science is to shape, control, and influence a given system, process, or phenomenon of interest. Starting with a model of the system, and once its validity is assured perhaps through experimental methods, one may then exploit knowledge and understanding of the model in order to affect the desired outcome. In the context of the models in this chapter, one is obviously interested in how to change or shape the discussion and opinion formation processes. We mention two different approaches.

One approach concerns how to empower an individual(s) during the opinion formation process. Recall from Corollary 1 that individual $i$'s power in setting the overall consensus value is given by $\gamma_i$, the $i$th entry of the left Perron–Frobenius eigenvector. Then, a natural problem to consider is how to adjust the network $\mathcal{G}$ to maximise $\gamma_i$ for some specific individual $i$, thus maximising their power. This may involve adjusting the weights $w_{ij}$, or adding or removing edges or nodes entirely. The vector $\gamma$ is sometimes referred to as the centrality vector, and $\gamma_i$ is the *eigenvector centrality* of node $i$ in the network $\mathcal{G}$. There is a large body of literature on eigenvector centrality, which has been studied as a network measure in the SNA literature and is also related to the PageRank metric used by the Google search engine (Gleich, 2015). Roughly speaking, $\gamma_i$ captures how well connected node $i$ is relative to other nodes, taking into account the overall network structure, which may indicate a potential to influence the overall network. Maximising an individual's influence can thus be cast as an optimisation problem of a node's centrality or PageRank (Amelkin & Singh, 2019; Avrachenkov & Litvak, 2006; Csáji et al., 2014; de Kerchove et al., 2008; Ye et al., 2017). Naturally, influence maximisation can be considered for models besides the standard DeGroot model. However, one must first identify an appropriate measure of influence, which is dependent on the model dynamics. For instance, the FJ model typically fails to reach a consensus, and hence the measure of influence is far more complex to compute compared with that of the DeGroot model, but can nonetheless be computed (Proskurnikov & Tempo, 2017).

A second approach considers introducing a limited number of zealots, opinion leaders, or stubborn extremists into a population in order to shift the opinions of the other (normal) individuals towards a desired target. We use the term "zealot" to broadly refer to an agent who is closed to social influence but able to influence other individuals[9], who may have a goal of steering the opinions of the wider population towards a certain objective value, and whose opinion updating rules may be pre-determined or designed to achieve the aforementioned goal. Some literature also name such agents "opinion leaders" and "stubborn extremists", and such agents may represent media, politicians, leaders and high-profile individuals (Quattrociocchi et al., 2014; Zhao et al., 2018). Some problems explore how two or more groups of zealots may compete against each other in trying to drive the population towards different desired opinion targets (Mao et al., 2021; Zhao et al., 2018). In the simplest implementation in the DeGroot or FJ models, one can represent a zealot by a stubborn agent $i$, with $w_{ii} = 1$, and typically having initial opinion $x_i(0)$ that is at one extreme end of the opinion spectrum; since $w_{ii} = 1$, the agent's opinion is constant (Acemoğlu et al., 2013; Ghaderi & Srikant, 2014; Ye, 2019). In more complex implementations, zealots may have $x_i(t+1) = u(t)$, where $u(t)$ is a time-varying control input (Dietrich et al., 2018), or $x_i(0)$ is selected through an optimisation procedure which aims to push the opinions of the wider population towards a certain value (Mao et al., 2021). Besides exploring different implementations of the agent dynamics for zealots, one can also explore their impact according to their position in the network (Amblard & Deffuant, 2004; Ye, 2019), the number of zealots present (Ye, 2019; Zhao et al., 2018), or other attributes of zealots (S. Chen et al., 2016). We end by pointing out that such studies do not typically consider how to introduce such individuals into the desired locations in the network, and focus only on establishing the effects of such individuals.

[9]In the network context, this implies there may be edges from other individuals to the agent but not vice versa.

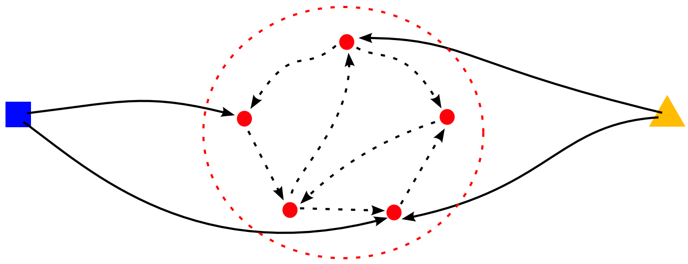


Fig. 8: Illustrative example of opinion leaders (media, politicians, leadership, celebrities). The network comprises of two opinion leaders (blue square, yellow triangle), and a group of regular individuals (red circles). While regular individuals can influence each other in a strongly connected subnetwork (red dashed circle), they cannot influence the leaders. On the other hand, the leaders can influence regular individuals, and may be trying to compete against one another for maximal influence in the network.

*2) Inferring network structure and influence:* As one may notice, Section II-B presents a variety of results that allow us to predict $x(t)$, viz. opinion formation outcomes through group discussions and repeated social interactions, given knowledge of the network as parametrised by $\mathcal{G}$ and $W$. We now deal with an *inverse problem*, seeking to learn of $W$ and $\mathcal{G}$ from observations of $x(t)$, perhaps obtained through surveys or estimated through data. A comprehensive treatment and survey of recent approaches dealing with inference for opinion formation models can be found in (Ravazzi et al., 2021), while Chapter 7 explores a statistical approach that does not assume the presence of an underlying dynamical system (e.g. the DeGroot model). So far, there have been limited study of models for capturing the processes behind network formation (e.g. how edges are created and what determines the $w_{ij}$ values) (Jia et al., 2015; Ye et al., 2018) — future work could drawn on inspirations from the literature around group formation and maintenance, see Chapter 9.

To further motivate the inverse problem, notice that Theorem 1 states that a consensus is achieved if the network is strongly connected. This conclusion holds even if one does not know the precise values of the influence weights $w_{ij}$, provided the assumption $\sum_{j=1}^{n} w_{ij} = 1$ holds. There are computationally fast and cheap algorithms for checking the strong connectivity of large-scale networks. However, if one were to desire more detailed knowledge, say of the final consensus value, then Corollary 1 states that we need to know the initial opinions $x(0)$ and the vector $\gamma^\top$. Stronger conclusions require more detailed knowledge, including the values for each entry of $W$ (the number of entries scales as $n^2$). Moreover, knowledge of $W$ and $\gamma^\top$ are also needed in many of the problems discussed in Section II-D1. One can immediately appreciate the nontrivial nature of estimating $W$ by noticing that given the final consensus value $\alpha$ and $x(0)$, the set of $W$ which satisfies $\gamma^\top x(0) = \alpha$ is an infinite set.

The problem of deriving the entries of the matrix $W$ can be approached in two distinct ways that are based on the availability of the observations $x(t)$ over a time period $0 \leq t \leq T$ (Ravazzi et al., 2021). If the opinions of all individuals have been sampled over $T$ rounds of conversation then the "persistent measurement" approach can be utilised. Given knowledge of the network structure, viz. the presence or absence of edges between nodes, and a sufficiently long set of consecutive observations $x(k), x(k+1), \ldots, x(k+T-1)$, it is possible to estimate $W$ (Ravazzi et al., 2021). The persistent measurement approach is only feasible in experimental studies such as (De et al., 2014; W.-X. Wang et al., 2011), where data is collected under controlled settings and the structure of the network is known.

Alternatively, the sporadic measurement approach is used when only intermittent observations of $x(t)$ are available, with infinite horizon (Ravazzi et al., 2018; Wai et al., 2016, 2018, 2019, 2020) and finite horizon (B. D. Anderson et al., 2020) being two popular approaches. The infinite horizon approach employs the FJ model under the assumption that the opinion formation process will converge to a steady state. It further assumes that there are $m$ independent topics being discussed, the initial opinions are available for each topic, and that we are able to sample the final opinions of all individuals as $t \to \infty$. Under the assumption of network sparsity and given enough topics ($m$ is sufficiently large) it is possible to recover the matrix $W$ through convex optimisation (Ravazzi et al., 2021). The finite horizon approach exploits dynamical properties and does not require the initial and final opinions to be known (Ravazzi et al., 2021). Rather, one assumes there is an observational model $z(t) = P(t)x(t)$, where $P(t)$ is the probability of observing $x(t)$ at time $t$. That is, opinions are measured intermittently through the vector $z(t)$, based on a probability distribution $P(t)$. The $z(t)$ measurements are then used to approximate a cross-correlation matrix $\sigma^{[l]}(\infty)$, which is in turn used to estimate the influence matrix $W$.

### *E. Empirical Studies*

In the following, we report different empirical studies that have been conducted, and discuss the inherent challenges that have limited this sector of the literature from developing in the same way as the literature covered so far. An important distinction must be made between *experimental* studies conducted in controlled laboratory environments, and those conducted on *observational* data gathered from real-world environments.

**Experimental studies** have been conducted by Noah Friedkin and his collaborators, focusing on the FJ model and its variants (N. Friedkin et al., 2019, 2021; N. E. Friedkin, Jia, & Bullo, 2016; N. E. Friedkin & Johnsen, 2011). Most of Friedkin's studies follow a similar format, although the topics and scenarios are varied. Individuals are separated into groups of $3$ to $4$ participants and asked to discuss a series of topics. The opinions of each participant prior to and post discussion are recorded. At the end of the discussion, each group member recorded how influential others were to their final decision, which informs $W$. Unfortunately, Friedkin's approach is limited to only consider small network sizes and face-to-face communication, which does not translate well into modelling online social media environments (we discuss this in Section IV). Friedkin has also explored discussion of multiple topics in sequence, viz. Section II-C5, finding that the influence weights change over the topic sequence and typically a single individual tends to dominate if the sequence is long enough (N. E. Friedkin & Bullo, 2017; N. E. Friedkin, Jia, & Bullo, 2016).

Other researchers have focused their efforts on other modifications of the DeGroot model (Becker et al., 2017; Kozitsin, 2020; Monti et al., 2020) or entirely novel model formulations (De et al., 2014, 2016). Becker et al. made a simple alteration to the classical DeGroot opinion dynamics model, preserving the weighted averaging mechanism, and then performed an online experimental study involving networks of $40$ participants, obtaining evidence that supported their model. Transitioning away from classical opinion dynamics models, De et al. developed a new linear model of opinion formation which does not assume that opinions are sampled at regular intervals (De et al., 2016). Besides convergence, De et al. further validated their theoretical findings through a series of experimental studies involving participants interacting through online predefined networks; the data suggested the proposed model outperformed existing models including the DeGroot model (De et al., 2016).

**Observational studies** provide a deeper insight into the applicability of an opinion dynamics model under real-world settings. Data is typically collected from social media platforms such as Facebook, Twitter or Reddit, and a natural language processing (NLP) model is used to derive real-valued opinion vectors based on the public text posts that users make (De et al., 2016; Kozitsin, 2020, 2021; Monti et al., 2020). Kozitsin applied the DeGroot model to observational data collected from over a million users on VKontakte in 2018, which is a popular Russian social media platform (Kozitsin, 2020). In addition to developing a method to estimate the network structure $W$ based on "friend connections", Kozitsin found that users had a high likelihood of moving their opinion towards that of their friends, supporting DeGroot's weighted averaging mechanism. Interestingly, it was also possible for a user's opinion to "skip" their friends' opinions and become more extreme, which is not accounted for under the DeGroot model. Another study, by Monti et al., examined whether the "backfire effect" as defined in Section II-C4 was present in online political discussions (Monti et al., 2020), by considering a bounded-confidence model with negative influence (see Section II-C4). Using archived Reddit posts from between 2008 and 2017 concerning political discussions, the authors showed the proposed model could capture some of the micro-level interactions within the subreddits, but were unable to find any instances of the "backfire effect".

In summary, experimental studies generally allow for more precise parameter estimations and provide a deeper insight into individual-level interactions during opinion formation (at least in small groups). However, one limitation is that performing experimental studies on networks with thousands of participants is infeasible, and hence most empirical studies involve face-to-face experiments (Friedkin's work) or small groups via an online platform (Becker et al., 2017). In contrast, observational studies on complex networks with partial opinion observations offer more"realistic" testing grounds for opinion dynamics models, and are conducted almost exclusively online via social media platforms. However, observational studies are challenging to conduct and compare against one another, as methods of extracting opinion vectors and network structures must be tailored to the data architecture of the specific social media platform. Opinion vectors are estimated using natural language processing (NLP) tools, but the accuracy of NLP may differ between discussion topics, further complicating comparison efforts.

## III. Models of Collective Decision-Making

In this section, we briefly explore models of decision-making in social networks. Much like opinion formation, the literature on decision-making models is vast and contributions flow from many different scientific communities. We present one popular framework, termed "binary decision-making", to allow comparison with models from Section II with opinion formation; this model has been studied in a range of contexts, including the evolution of social norms and conventions (Peyton Young, 1993, 2015; Ye & Zino, 2024; Ye, Zino, Mlakar, et al., 2021), and the diffusion of social innovations (Kreindler & Young, 2014; Montanari & Saberi, 2010; Peyton Young, 2009; Zino et al., 2020).

### *A. The basic model dynamics*

The underlying framework comes from game theory, which has a long tradition in being used to capture decision-making for interacting rational or selfish actors (Peyton Young, 2001; Von Neumann & Morgenstern, 1944). In a population of $n$ individuals indexed by the set $\mathcal{V} = \{1, \dots, n\}$, each individual can select between two strategies from the set $\mathcal{S} = \{0, 1\}$. In the context of social innovation, 0 and 1 may represent the status quo and innovation, respectively. In the context of social conventions, 0 and 1 may represent two alternative conventions, e.g. handshake vs. elbow-bump as a form of greeting. Similar to opinion formation models, we may assume a discrete-time process occurring at time instants $t = 0, 1, \dots$, and the presence of an underlying graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, W)$ governing the interactions among individuals, with the same definition of nodes and edge set as in Section II-B. For the sake of simplicity, we are going to assume that $w_{ij} = 1$ or $w_{ij} = 0$ (Riehl et al., 2018), in contrast to Section II-B (having non-binary valued weights is a straightforward extension, see e.g. Zino et al. (2020)).

For each individual $i \in \mathcal{V}$, let $x_i(t) \in \mathcal{S}$ represent their strategy at time $t$, and

$$x_{-i}(t) = (x_1(t), \dots, x_{i-1}(t), x_{i+1}(t), x_n(t))$$

represent the strategies of all individuals except individual $i$. In order to present the dynamics, we first formulate the game. Each player has a payoff function $\pi_i(x_i, x_{-i})$ that determines the reward for selecting strategy $x_i \in \mathcal{S}$ given that others have strategy $x_{-i}$. The reward can reflect both explicit (e.g. monetary) and implicit (e.g. social conformity or benefits) value from selecting a strategy. The payoff is

$$\pi_i(x_i, x_{-i}) = \sum_{j=1}^{n} w_{ij} \begin{bmatrix} x_j, & 1 - x_j \end{bmatrix} \begin{bmatrix} 1+\alpha_i & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_i \\ 1 - x_i \end{bmatrix}, \tag{10}$$

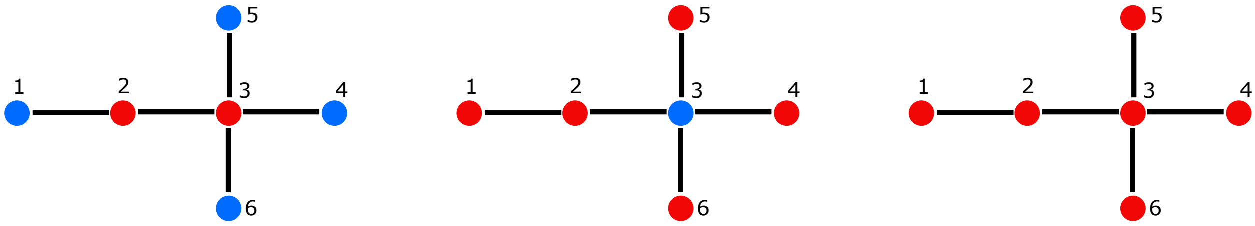
where $\alpha_i \in (-1, \infty)$ is called the "relative advantage" or "evolutionary advantage" of strategy 1 with respect to strategy 0, from the perspective of individual $i$. This payoff function reflects the fact that individual $i$ enters into a pairwise coordination game with each of their neighbours $j$: individual $i$ will receive a unit reward for coordinating their strategy with a neighbour $j$ playing strategy 0, and a unit reward adjusted by $\alpha_i$ for coordinating with a neighbour $j$ playing strategy 1. This model is thus sometimes referred to as "coordinating games on networks" (Jackson & Zenou, 2015).

At each time $t$, an individual may revise their strategy following a particular *update rule*, which can be carried out synchronously for all individuals or asynchronously for a subset of individuals (see Section II-C1) (Ramazi et al., 2016; Riehl et al., 2018; Zino et al., 2020). Different update rules can be studied depending on context and to capture different decision-making approaches, and one can then compare the emergent population outcomes. Popular rules include "imitation" and "best-response" (Riehl et al., 2018). Under imitation updating, an individual copies the strategy of their neighbour (in the network sense) who is receiving the largest reward, similar to a person deciding to invest in a stock that is rewarding their friend with great monetary return. Under best-response updating, an individual selects the strategy that offers the highest reward. For binary strategies under the payoff structure in Eq. (10), this yields $x_i(t+1) = 1$ if $\pi_i(1, x_{-i}(t)) > \pi_i(0, x_{-i}(t))$ and $x_i(t+1) = 0$ if $\pi_i(1, x_{-i}(t)) < \pi_i(0, x_{-i}(t))$ (one can resolve the special case of $\pi_i(1, x_{-i}(t)) = \pi_i(0, x_{-i}(t))$ in a variety of ways, for instance the individual does not change strategy). For a given $x_{-i}(t)$, straightforward calculations reveal that $\pi_i(1, x_{-i}(t)) > \pi_i(0, x_{-i}(t))$ if and only if

$$\frac{1}{d_i} \sum_{j=1}^{n} w_{ij} x_j(t) > \frac{1}{2 + \alpha_i} \tag{11}$$

where $d_i = \sum_{j=1}^{n} w_{ij}$ is the number of neighbours of individual $i$. A *noisy* best-response is commonly considered, with a variety of formulations, but in all such formulations each individual has a non-zero probability of selecting the strategy which gives the lesser reward (Blume, 1995; Kreindler & Young, 2014; Montanari & Saberi, 2010; Ye, Zino, Mlakar, et al., 2021).

To understand what Eq. (11) is telling us, suppose first that $\alpha_i = 0$ (meaning the two strategies provide equal payoff in the coordination game). An illustrative example is provided in Figure 9. Because $\frac{1}{d_i} \sum_{j=1}^{n} w_{ij} x_j(t)$ is the fraction of individual $i$'s neighbours selecting strategy 1, it follows that $x_i(t+1)$ will be the strategy that is played by the majority of the neighbours at time $t$. In other words, individual $i$ will coordinate and conform to the most prevalent strategy among their peers. This yields threshold-type dynamics, where an individual will stick to their current strategy until a threshold, defined in Eq. (11), is exceeded due to changes in the strategy of their neighbours. It is also not difficult to notice that best-response updating in this formulation captures an averaging process, but the average used is the *median* as opposed to the *mean* in Section II. We note that the DeGroot model (and others such as the FJ and Altafini models) has a game-theoretic interpretation, with Eq. (1) being the best-response update for the payoff function $\pi_i(s, x_{-i}(t)) = -\sum_{j=1}^{n} w_{ij}(s - x_j(t))^2$ (Ghaderi & Srikant, 2014; Marden et al., 2009). A median weighted averaging opinion dynamics model has been recently proposed (Mei, Bullo, et al., 2022; Mei et al., 2024). When $\alpha_i > 0$, then the value on the right of Eq. (11) is *smaller* than $1/2$, which means that the threshold for individual $i$ to select $x_i(t+1)$ is lowered: fewer neighbours are needed to be playing strategy 1 at time $t$ for individual $i$ to switch. It is therefore clear that $\alpha_i > 0$ and $\alpha_i < 0$ represents an advantage and disadvantage, respectively, that strategy 1 has over strategy 0, which lowers and raises the threshold.




(a) time $t$ (b) time $t+1$ (c) time $t+2$

Fig. 9: Illustrative example of binary coordination decision-making on a network, where nodes can either select strategy 0 (blue) or strategy 1 (red). In this implementation, $\alpha_i = 0$ in Eq. (10) for all individuals $i$, and at each time step, each individual revises their strategy according to a best-response strategy, which with $\alpha_i = 0$ amounts to selecting the strategy adopted by the majority of your neighbours – if payoffs are equal, the individual will not change strategy. (a) Nodes 1, 4, 5, and 6 each have a single neighbour, who are playing red, and hence will choose to red at $t+1$. Node 2 has one neighbour of each strategy, and its strategy will remain unchanged. Node 3 has three neighbours playing blue and one red, and will switch to blue at $t+1$. (b) Node 2 again will not change its strategy. All neighbours of Node 3 are playing red, and hence it will switch to red at $t+2$. (c) In this example, after two time steps, individuals have collectively adopted the same strategy, which was in fact the minority strategy at the start. This example outcome is due to the network structure and initial strategy choices, and is not always guaranteed.

### *B. Applications and extensions*

Naturally, it is of interest to explore whether or not the dynamics converge, viz. whether individuals eventually settle on a decision they are content with (Ramazi et al., 2016; Riehl et al., 2018). Rather than detail convergence results, which boasts an extensive literature, we conclude the section by discussing the different applications contexts, and a few extensions.

One of the most prominent applications of the model is to study the emergence of collective behaviour and decision-making at the population level through individual-level coordination, including the diffusion of *social innovation* (Kreindler & Young, 2014; Montanari & Saberi, 2010; Peyton Young, 2009; Ye, Zino, Mlakar, et al., 2021). In this scenario, we assume that at time $t = 0$, all individuals are selecting strategy 0, which represents the *status quo* behaviour, product, social convention, norm etc. An *alternative* is introduced, represented by strategy 1, and one is interested in how the alternative spreads, including how many individuals eventually adopt the alternative and how quickly the diffusion process occurs. Note that "innovation" is an overarching term for something that is newly introduced to the population, and does not necessarily mean that the alternative has an advantage $\alpha_i > 0$ over the status quo (Rogers, 2003). A common mechanism to drive diffusion is to introduce into the population a committed minority who stubbornly select the alternative strategy 1 for all time (sometimes referred to as zealots, see Section II-D1) (Centola et al., 2018; Ye, Zino, Mlakar, et al., 2021), and one may wish to determine the relationship between the size of the committed minority and how widely the alternative diffuses. A second approach is to investigate the magnitude of the advantage $\alpha_i > 0$, and how it determines the diffusion of the alternative (Kreindler & Young, 2014; Montanari & Saberi, 2010; Zino et al., 2022).

Much like the many different opinion formation models that evolved from the DeGroot model, there are many works that expand on the basic collective decision-making model presented above. We mention just a few, to highlight some important extensions.

- Notice that in the formulation in Section III-A allows an individual to switch between the two strategies at each time instant. While this is suitable for capturing an individual selecting between handshakes and bowing as a form of greeting, it is not suitable for capturing an individual deciding whether to install solar panels on their home (alternative) or not (status quo). Put simply, there are scenarios in which an individual that switches from $x_i = 0$ to $x_i = 1$ are unable to switch back. This variant of the model has been extensively studied, including recently as a *complex contagion* model (Guilbeault et al., 2018), and is also known as the Linear Threshold Model (LTM) (Centola & Macy, 2007; Granovetter, 1978; Kempe et al., 2003; Macy & Evtushenko, 2020).
- The matrix in Eq. (10) captures a standard 2-player 2-strategy (2 by 2) symmetric *coordination game*. A variety of other classic 2 by 2 games such as prisoner's dilemma, anti-coordination, Snowdrift, Stag-hunt, etc., can instead be considered by adjusting the values of the four entries of the matrix (Riehl et al., 2018). This allows to capture different scenarios and pairwise interactions. For instance, an anti-coordination game might capture two potential evacuation routes (strategy 0 and 1) away from a disaster, and one would typically aim to *avoid* the strategy (route) that is more popular to prevent congestion. This stream of literature spills over into a large body of work on evolutionary dynamics and evolutionary game theory (Hofbauer & Sigmund, 1998; Riehl et al., 2018), concerned with how different behaviours, norms, genetic traits, subspecies, etc. persist and evolve in social or biological populations.
- Finally, we remark that the game-theoretic formulation has several advantages as it separates the payoff from the update rule. Different update rules can reflect different approaches to decision-making, while it is straightforward to incorporate additional strategies, allowing to study 3 or more choices to better reflect real-world decision-making. One can also incorporate additional factors into the decision-making by adding additional terms to the payoff. The recent work Ye, Zino, Mlakar, et al. (2021) on social conventions incorporated status quo bias (a person's reluctance to change from whatever they are currently doing) (Samuelson & Zeckhauser, 1988) and dynamic norms (a person's tendency to notice and follow trends even when something has not reached the mainstream) (Noelle-Neumann, 1993; Sparkman & Walton, 2017), while Ye, Zino, Rizzo, and Cao (2021) explored decisions on adopting self-protective behaviours during a pandemic by incorporating accumulating social-economic fatigue, fear of infection, and external government interventions.

As noted at the start of this Section III, game-theoretic models can be used to capture social norms and conventions. These models could therefore be used to study the formation, maintenance, and change of social norms, as detailed in Chapter 10. This could be done in a war game setting to determine the success of particular spreading or evolutionary mechanisms, whether collective action or a common social identify will emerge, and whether population-level cultural evolution will occur.

## IV. Discussions

Arriving at the final section of this chapter, we have introduced models of opinion formation and decision-making via social influence, and then presented a range of extensions and applications. We finish by providing a broad discussion on the future research, in particular the key directions that should be pursued in order to apply such models to grey zone environments and societal threats faced by Australia in the $21^{st}$ Century.

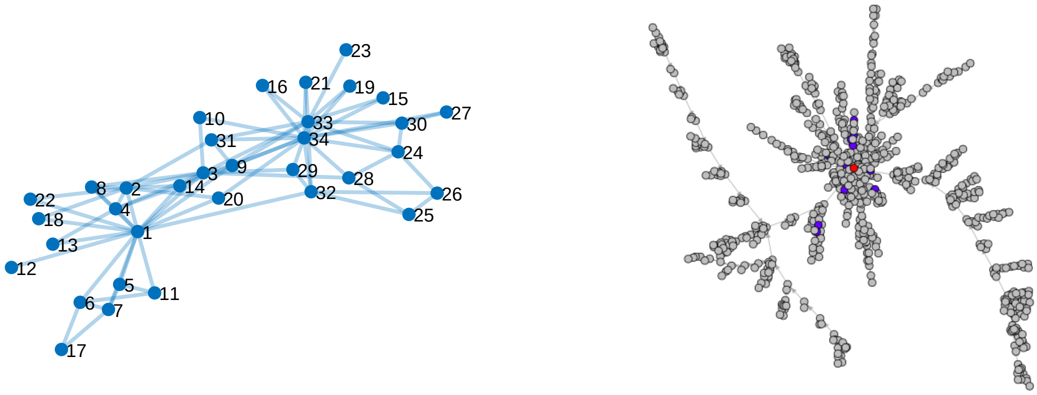
### *A. Advances in Modelling*

Since the work of French Jr., significant attention has been devoted to advancing the basic DeGroot model by introducing additional agent and network-level features that aim to describe and capture known phenomena from the sociological and psychological literature, resulting in better representation of real-world interaction and discussion. Besides introducing more novel sophisticated features, researchers have also combined $X$ and $Y$ features of different existing models and studied the resulting model. The result is in an explosion of models that are each variants or generalisations of existing ones. This has in part been aided by advances in computational methods, allowing for rapid simulation of complex nonlinear models. However, models have reached a point in complexity and maturity where the landscape is saturated. It must be pointed out that just because a phenomenon is widely reported in the literature, does not imply said phenomenon necessarily occurs during opinion formation, which is a specific social process among many. Therefore, further model features should be introduced only if accompanied by rigorous and direct experimental evidence of their presence *during opinion formation*. For instance, while an individual's sensitivity to observing growing trends is well documented as a general social psychological phenomenon (Noelle-Neumann, 1993; Sparkman & Walton, 2017), one should also obtain strong evidence such sensitivity is present during opinion formation or group decision-making (Ye, Zino, Mlakar, et al., 2021). A related challenge is to recognise that the models of this chapter assume precise mechanisms are at play during opinion formation and decision-making, but as elaborated upon in Chapter 4, the way individuals process information (such as shared opinions) is complex and heavily influenced by contextual factors — such considerations have typically been overlooked in the opinion modelling literature. Keeping these issues in mind will help limit model advances to those which incorporate features that are indispensable to capturing real-world opinion phenomena.

Perhaps most importantly, we must recognise that the fundamental DeGroot model arose during a time when people conversed face-to-face, and discussions occurred through a clear social network structure, such as a company board or local political chapter. The presence of clear social ties between individuals is fundamentally related to the social influence weights $w_{ij}$ in Eq. (1). In the digital age, online social media platforms such as Twitter, Reddit, and Facebook form a fundamental part of the social fabric for each person. We receive information from sources and interact with people across a wider spectrum simply inconceivable even 20 years ago, and much of it is filtered and determined by recommender systems (Jannach, 2010). Online discussions can involve actors lacking in permanent identification (Twitter users can change their handle/username at will), and interactions are often fleeting with no expectation of future repeated discussion. Users may be unaware or unable to remember past interactions with each other, and permanent social ties are the exception rather than the rule. Figure 10b presents an example of the differences between a typical social network and an online discussion network. One key direction is the sustained development of models of opinion formation which accurately reflect the key characteristics of online discourse, the data available for collection, and represents the social media architectures that define how interactions and discussions occur (Rossi et al., 2021; Sprenger et al., 2024).

### B. Advances in Application and Theory

In any scientific pursuit, there is a natural progression of knowledge. First, scientists aim to describe the phenomena of interest, often through building theory and mathematical models. Once models are validated, typically through experimental methods, a concrete test of the model and theory is the ability to apply the framework to real-world scenarios, and introduce control and interventions to shape the phenomena. In Section II-D, we explored several applications of opinion formation models, namely shaping the discussion outcomes, impact of zealots and extremists, and inferring network structure. We believe these efforts need to expand, in parallel with the modelling advances discussed above, in order to build a deeper theoretical basis for applying these models to address future problems in grey zone environments.




(a) (b)

Fig. 10: Illustrative examples of social networks. (a) Zachary's Karate Club: a traditional social network of a children's karate school, with edges between nodes (individuals) reflecting social ties. (b) An online discussion network on Reddit, where nodes represent comments posted by a dynamically changing group of users, links denoting replies between comments. Zachary's network reconstructed from data in Zachary (1977), Reddit network supplied by R. Ackland.

Minority influence has long been of interest in social psychology (Moscovici et al., 1969), as scientists seek to understand how a small vocal minority can significantly shape the discourse and movements of entire populations. Recent experimental and theoretical work using opinion formation and collective decision-making models have highlighted how a small committed minority can overturn the status quo (Centola et al., 2018; Ye, Zino, Mlakar, et al., 2021). Moreover, recent insights from dynamic norms (Sparkman & Walton, 2017; Ye, Zino, Mlakar, et al., 2021) and past literature on the "spiral of silence" (Noelle-Neumann, 1993) show that people are perceptive to trends. This suggests it is not only the size of the committed minority, but also the rate at which they appear, which can determine their long-term success. Network inference approaches must continued to be pursued, as the theory and methods are still underdeveloped relative to the other parts of the literature. However, these approaches should be carefully formulated keeping in mind that the data-rich environments of online social media platforms may provide data that do not map one-to-one with the existing models, as we saw in Section II-E. By developing models that more accurately capture online environments, as described in Section IV-A, network inference approaches can better exploit the data available. It would also be of interest to explore the connection the information transmission models discussed in Chapter 5, and opinion formation and decision-making models, noting that the models discussed in this chapter do not explicitly consider 'flow' or 'transmission' over social networks but rather posits that social influence leads to change of opinions or decisions.

In much of the existing literature, problems exploring zealots and influence maximisation often make the unrealistic assumption that the influence is constant, e.g. zealots display a constant opinion that is unchanging over time. An important future direction is to explore *intelligent* agents who can respond to changes in the network and population, perhaps implementing an optimal opinion trajectory to steer the rest of the individuals towards a desired outcome or learning the best trajectory by repeatedly stimulating the population and observing the response. The latter may draw on the growing literature on *learning* for autonomous systems. Problems involving a group of intelligent agents who either coordinate or compete with one another should be pursued, with great applicability to wargames. A natural extension to introducing dynamic intelligent influence agents is to develop methods for identifying and detecting such agents. While static zealots can be easily identified in a model, coordinated inauthentic activities can be significantly harder to detect in reality (Smith et al., 2021; Subrahmanian et al., 2016). One promising direction is to explore the applicability of the literature on security and attack-detection of cyber-physical systems (Etesami & Başar, 2019; Humayed et al., 2017; Pasqualetti et al., 2013). Here, control-theoretic techniques are used to explore whether an attacker injecting malicious inputs into a cyber-based controller or feeding false sensor signals can damage the physical system (e.g. a processing plant, an electric generator), and whether such actions can be detected or not. By advancing the applications of existing models, this will allow one to study problems that more closely resemble real-time influence operations in grey zone environments.

Finally, we highlight the fact that this chapter has considered a population of interacting individuals as a closed and isolated system. However, as evident from Chapters 1, 11, 12 and 13, individuals (or groups of individuals) do not exist in a vacuum but rather interact with, contribute to, and ultimately help to shape, the institutions and information environments that form the human-made environments that every individual is embedded in. How to incorporate institutions and various aspects of the information environment into the models of this chapter (and whether such incorporation is necessary to advance the application of such models) are problems that have not been addressed — there is significant room to explore such questions in collaboration with researchers and practitioners from the social sciences and other disciplines.

### *C. Advancements in Deployment and Validation*

Great strides have been made in the theoretical development and application of opinion formation models; however, there has been little movement in translating these advancements towards implementation in practice and development of capabilities at higher Technology Readiness Levels. To stoke interest in such development, we believe it necessary to implement model validation methods, standard in other scientific domains, along with a suite of software/libraries aimed at allowing non-experts to visualise and utilise the existing collection of opinion dynamics models. The proposed approach is inspired by the field of machine intelligence, which has rapidly grown in popularity and scale over the past few decades, and has been successfully implemented as a maturing technology in a broad range of domains.

Transparent model validation methods are key to informing the direction of research and allowing comparisons between existing models. The field of machine learning utilises standard testing datasets (E. Anderson, 1936; J. Deng et al., 2009; L. Deng, 2012), which allows the *fair comparison* of the performance of subsequent models. Previous studies have compared the accuracy of opinion dynamics models (De et al., 2014, 2016), but used differing datasets and time periods, which does not allow for an accurate evaluation of model performance. Additionally, the non-standardised use of natural language processing (NLP) makes comparing models difficult. It is common practice to first implement an NLP model to estimate opinion vectors, which are real-valued, prior to studying the opinion dynamics model. Thus, it is possible for the same opinion dynamics model to perform differently on identical datasets, because of the difference in estimated opinion vectors when using different NLP models. Unfortunately, it is not possible to solve this problem by enforcing a homogenised method of opinion vector estimation, as different opinion topics and/or datasets require varying NLP approaches. One potential solution would be to create standardised opinion vector datasets based on collected data, allowing a fairer comparison of models and greater insight into the performance of different studies.

The metrics used in measuring the performance of opinion dynamics models should also be considered in greater detail, with efforts made to create some standardised metrics. Opinion dynamics model are dual purpose in that they are able to model the transition of group opinions over time, and that they can be used to infer the structure of the underlying social network. The distinction between the two modalities should be addressed by the accuracy metrics used to measure the performance of opinion dynamics models. One solution could be to use a weighted average of the accuracy of the network inference, and the opinion vector transitions.

It is important from an implementation perspective to understand how and what phenomena a model captures; however, it is not necessarily beneficial to understand the underlying inference techniques. To this end, we suggest that non-experts, such as wargamers or policymakers wishing to explore specific scenarios, are provided with the ability to implement opinion dynamics models. A suite of software packages/libraries should be developed to provide end-to-end pipelines for opinion vector estimation from collected data, modelling of different scenarios, and exploration of applications problems such as influence operations, interventions and grey zone activities. Additionally, visualisation tools should be provided to aid in understanding the models. The models discussed in this chapter have multiple features of interest which need to be visualised together, such as the network structure and temporal evolution of the discussion process unfolding over it. Compare for instance the temporal-only visualisation in Figure 2 against the temporal-network visualisation in Figures 3 to 5. This is a similar approach taken by the field of machine learning, which has developed tools such as Tensorflow, Scikit-Learn, and Pytorch (Martín Abadi et al., 2015; Paszke et al., 2019; Pedregosa et al., 2011) to allow non-experts to implement complex machine learning models.

Finally, the emergence of powerful and mature Large Language Models (LLMs), and more broadly generative Artificial Intelligence (AI) has already profoundly changed many scientific disciplines, and societal constructs and institutions. It is natural to consider how one can exploit LLMs and generative AI within the field of opinion formation and decision-making modelling. For instance, there are now attempts to use LLM to help program agents who process and generate information in more sophisticated and human-like ways (Lu et al., 2024; Perez et al., 2024), when compared to the abstract dynamics such as in Eq. (1). Given the discussions on challenges in obtaining and evaluating real-world data, LLMs may offer substantial improvements in converting text messages obtained from social media, into the opinions of the form in Section II, and to using generative AI to produce synthetic data that is representative of real-world data (DeBuse & Warnick, 2024).

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